

EF-DOC-008

EventFlow Application Services & Domain Implementation Design

Version 1.0 - Approved Baseline

Sprint	Sprint 6
Input baseline	v0.6.0-security
Service packages	SVC-001 through SVC-016 - 16/16 design PASS
Refinements	S6-R01 through S6-R75 - accepted
Integrated validation	IV-001 through IV-020 - 20/20 design PASS
Blocking contradictions	None

1. Consolidation Verdict

Sprint 6 is internally consistent and implementation-ready at the design level. The service architecture preserves the approved Sprint 3 database, Sprint 4 API/application-service, and Sprint 5 security baselines. Identified schema/API/catalogue changes are controlled forward extensions. No unresolved authoritative ownership, transaction, authorization, dependency-cycle, security, privacy, or event-day-operability contradiction was identified.

2. Implementation Architecture Invariants

- **IMP-INV-01:** Business mutations occur only through authoritative Application Services.
- **IMP-INV-02:** Controllers, WordPress hooks and workers do not directly perform domain persistence.
- **IMP-INV-03:** Repositories do not authorize callers or own external side effects.
- **IMP-INV-04:** Every Event-scoped mutation derives Event scope server-side and validates resource ownership.
- **IMP-INV-05:** Material business operations use explicit transaction boundaries.
- **IMP-INV-06:** External network calls do not execute inside authoritative business DB transactions.
- **IMP-INV-07:** Required Audit records commit atomically with the mutations they protect.
- **IMP-INV-08:** Background jobs assume at-least-once execution and are idempotent/restart-safe.
- **IMP-INV-09:** Raw reusable bearer credentials are never persisted as ordinary business data.
- **IMP-INV-10:** Optimized projections/read models do not bypass authoritative mutation services.
- **IMP-INV-11:** Unknown principal/provider/job/parser/report/policy identifiers fail closed.
- **IMP-INV-12:** Earlier approved baselines change only through controlled refinements/migrations, never silent rewrites.

3. Service Package Baseline

SVC-001 - Application Bootstrap & Dependency Composition

Single composition root; deterministic boot; explicit dependency injection; schema/runtime compatibility gate; thin WordPress adapters.

SVC-002 - Event Lifecycle Service

Atomic Event creation with default configuration and primary owner; explicit lifecycle commands/readiness; venue snapshotting; archive/restore semantics.

SVC-003 - Membership & Authorization Integration

Current server-side membership authority; capability/delegation policy; owner continuity; primary-owner transfer; request-authoritative authorization.

SVC-004 - Invitation & Guest-Access Service

Invitation lifecycle; return-once high-entropy credentials; guest-link indirection; server guest sessions; token-version invalidation; CSRF integration.

SVC-005 - Attendee & RSVP Service

Atomic complete-state RSVP reconciliation; response revision; capacity locking; attendee admin; primary-attendee continuity; seating-group synchronization.

SVC-006 - Import Pipeline Service

Secure staged CSV/XLSX pipeline; mapping/validation/dry-run; logical application units; domain-service application; restartable workers; retention.

SVC-007 - Seating & Recommendation Service

Flexible tables/seats/groups; authoritative manual assignment; deterministic locking; constraint classes; advisory versioned recommendations; stale detection.

SVC-008 - Reception & Check-In Service

Least-privilege reception projection; immutable attendee-level check-in/reversal actions; atomic bulk check-in; stations; mandatory idempotency.

SVC-009 - Communications & Campaign Service

Immutable template versions; explicit campaign purpose/audience mode; execution-time audience freeze; immutable Message snapshots; async provider dispatch.

SVC-010 - Provider & Webhook Processing

Provider-specific auth; durable-before-ack webhook capture; PDM-017A dedupe; normalization/correlation; out-of-order evidence; async reconciliation.

SVC-011 - Audit, Idempotency & Transaction Infrastructure

Shared transaction/idempotency/audit model; typed audit; tamper-evident chains; retry policy; durable async intent; at-least-once correctness.

SVC-012 - Background Jobs & Scheduling

Durable typed jobs; leases/heartbeats; priorities; bounded retries/dead-letter; logical dedupe; payload versioning; scheduler reconciliation.

SVC-013 - Reporting & Export Service

Event-scoped report projections; separate PII export capability; protected expiring artifacts; streaming generation; download reauthorization; export audit.

SVC-014 - Privacy & Retention Operations

Policy-driven retention; holds; restart-safe anonymization; privacy tombstones; post-restore reconciliation; forward-completion privacy recovery.

SVC-015 - Error Handling & Observability Integration

Single error catalogue; typed public details; centralized redaction; structured logs/metrics; health vs readiness; sanitized diagnostics.

SVC-016 - Service Integration & Implementation Validation

Cross-service reconciliation; dependency-cycle controls; transaction/authorization integration; 20 integrated workflows; baseline extension register.

4. Cross-Cutting Integration Model

Dependency direction

WordPress/REST/CLI/workers -> Application Services -> shared application infrastructure/domain policies -> repositories -> database/provider/host adapters. Concrete cross-domain cycles are prohibited; narrow ports/orchestrators resolve collaboration.

Transaction ownership

Application Services own authoritative business transactions. Repositories participate without independent commit. External provider/network calls are outside DB transactions. Large workflows use bounded restart-safe units.

Authorization

PrincipalContext + current server-side Event membership/global capability + lifecycle overlay + resource/Event scope + operation-specific policy. UI state and client claims are never authoritative.

Asynchronous work

Durable typed Jobs are authoritative; schedulers only trigger. Workers assume at-least-once execution and use leases, logical dedupe, idempotency, bounded retry, dead-letter and reconciliation.

Guest security

Invitation and guest-link/session bearer credentials are high entropy and stored only as digests. Rotation/revocation invalidates prior token versions. Message history never stores the underlying long-lived Invitation token.

Audit/observability

Required business/security Audit commits with protected mutations. Audit payloads are typed/minimized and may be chained for tamper evidence. Logs/metrics/diagnostics are separate, redacted, structured and non-authoritative.

Privacy/recovery

Archive, uninstall, user deletion and privacy purge are distinct. Privacy workflows are restart-safe and forward-completing, leave durable non-PII state markers, and gate post-restore readiness until reconciliation completes.

5. Sprint 6 Refinement Register

- **S6-R01:** Runtime schema compatibility gate.
- **S6-R02:** Idempotent Event creation.
- **S6-R03:** Request-authoritative membership authorization.

- **S6-R04:** Revoked Invitation reactivation requires a new credential.
- **S6-R05:** Supporting schema for server guest sessions and per-message guest-link credentials.
- **S6-R06:** Persisted RSVP response revision.
- **S6-R07:** Primary attendee continuity.
- **S6-R08:** Idempotent organizer attendee creation.
- **S6-R09:** Import row/application-unit idempotency.
- **S6-R10:** Logical import-unit atomicity.
- **S6-R11:** Imported Invitation credential handling without raw-token persistence/export.
- **S6-R12:** Import worker lease.
- **S6-R13:** Deterministic seating lock order.
- **S6-R14:** Constraint override classes.
- **S6-R15:** Recommendation algorithm versioning.
- **S6-R16:** One active recommendation job per Event.
- **S6-R17:** Planning assignment lock.
- **S6-R18:** Recommendation reproducibility seed.
- **S6-R19:** Mandatory check-in and bulk-check-in idempotency.
- **S6-R20:** Dedicated reception lookup identifiers for QR/barcodes.
- **S6-R21:** Optional rebuildable check-in current-state projection.
- **S6-R22:** Template preview uses non-functional credentials.
- **S6-R23:** Explicit dynamic versus snapshot audience mode.
- **S6-R24:** Campaign-recipient Message idempotency.
- **S6-R25:** Ambiguous provider outcome state.
- **S6-R26:** Destination correction creates a new Message.
- **S6-R27:** Explicit Campaign purpose.
- **S6-R28:** Campaign execution immutability.
- **S6-R29:** Durable webhook acknowledgement.
- **S6-R30:** Provider dedupe algorithm versioning.
- **S6-R31:** Unmatched ProviderEvent reconciliation.
- **S6-R32:** Conflicting provider evidence preservation.
- **S6-R33:** Delivery-attempt correlation metadata.
- **S6-R34:** Engagement tracking opt-in design.
- **S6-R35:** Webhook ingestion/processing separation.
- **S6-R36:** Provider-event processing idempotency.
- **S6-R37:** Provider correlation namespace.
- **S6-R38:** Explicit nested transaction policy.
- **S6-R39:** Explicit transactional retry eligibility.
- **S6-R40:** Sensitive return-once idempotency semantics.
- **S6-R41:** Transactional idempotency completion.
- **S6-R42:** Typed audit event catalogue.
- **S6-R43:** Versioned audit canonicalization.
- **S6-R44:** Durable asynchronous intent reconciliation.
- **S6-R45:** No distributed exactly-once assumption.
- **S6-R46:** Idempotency execution lease.
- **S6-R47:** Shared background-job lease model.
- **S6-R48:** Explicit dead-letter handling.
- **S6-R49:** Durable logical job deduplication.
- **S6-R50:** Worker schema compatibility gate.
- **S6-R51:** Versioned job payload contracts.
- **S6-R52:** No executable job serialization.

- **S6-R53:** Explicit sensitive-export purpose.
- **S6-R54:** Explicit export temporal semantics.
- **S6-R55:** Reauthorize sensitive export download.
- **S6-R56:** Bounded export concurrency.
- **S6-R57:** Atomic export artifact publication.
- **S6-R58:** No long-lived export transaction.
- **S6-R59:** Explicit Event Export resource.
- **S6-R60:** Privacy-safe immutable audit strategy.
- **S6-R61:** Retention/legal hold.
- **S6-R62:** Restart-safe privacy orchestration.
- **S6-R63:** Durable privacy tombstone/state marker.
- **S6-R64:** Privacy reconciliation readiness gate.
- **S6-R65:** Privacy action invalidates affected temporary exports.
- **S6-R66:** Versioned privacy/retention policy decisions.
- **S6-R67:** Privacy failure uses forward completion, not PII rollback.
- **S6-R68:** Privacy as explicit resource; routine retention execution non-public.
- **S6-R69:** Typed public error details.
- **S6-R70:** Central observability redaction policy.
- **S6-R71:** Low-cardinality metrics discipline.
- **S6-R72:** Sanitized diagnostic export.
- **S6-R73:** Single authoritative error catalogue.
- **S6-R74:** Cross-domain collaboration through narrow ports.
- **S6-R75:** Optimized read models remain non-authoritative for mutation.

6. Integrated Implementation Validation

- **IV-001 - Create and activate Event: PASS.** Event + default configuration + primary owner; readiness; venue snapshot; idempotent creation.
- **IV-002 - Invite guest securely: PASS.** Return-once Invitation token; per-Message guest link; guest session; no raw-token persistence.
- **IV-003 - Guest RSVP amendment: PASS.** Complete-state reconciliation, capacity, response revision, invitation-group sync.
- **IV-004 - Concurrent RSVP capacity: PASS.** Invitation locking prevents capacity overrun.
- **IV-005 - Organizer attendee correction: PASS.** Attendee edit without unintended seating/check-in mutation.
- **IV-006 - Seating preparation: PASS.** Variable tables/seats/groups; readiness and accessibility/capacity checks.
- **IV-007 - Recommendation workflow: PASS.** Snapshot fingerprint; manual change causes stale apply rejection.
- **IV-008 - Host manual override: PASS.** Authorized social-constraint override with reason; non-overrideable accessibility constraint remains blocked.
- **IV-009 - Reception event-day flow: PASS.** Least-privilege search and atomic family/selected bulk check-in.
- **IV-010 - Check-in reversal: PASS.** Elevated authority and reason; immutable reversal.
- **IV-011 - Scheduled Campaign: PASS.** Dynamic execution-time audience; immutable Message freeze; guest-link issuance.
- **IV-012 - Provider outage: PASS.** Circuit breaker/backoff; unrelated RSVP/seating/check-in remain available.
- **IV-013 - Provider webhook replay: PASS.** Authenticated durable capture; PDM-017A dedupe; no duplicate Message transition.
- **IV-014 - Ambiguous provider result: PASS.** No blind resend; bounded reconciliation.
- **IV-015 - Large import: PASS.** Secure parse, dry-run, logical units, worker crash/resume without duplicate domain mutations.
- **IV-016 - PII export: PASS.** Purpose, async protected artifact, download reauthorization, audit, expiry.

- **IV-017 - Privacy action: PASS.** Hold check, credential revocation, PII minimization, export invalidation, tombstone.
- **IV-018 - Backup restoration: PASS.** Readiness block until privacy reconciliation reapplies prior privacy state.
- **IV-019 - Scheduler failure: PASS.** Durable pending work survives scheduler failure and is re-enqueued.
- **IV-020 - Archive Event: PASS.** Writes/check-in denied; reporting allowed; late provider evidence retained; restore to completed only.

7. Baseline Reconciliation

Area	Result	Reconciliation
Sprint 3 Database	PASS WITH CONTROLLED EXTENSIONS	Core physical model remains frozen. New support entities/metadata are introduced only through future versioned migrations.
Sprint 4 API	PASS WITH CONTROLLED ADDITIONS	Existing v1 endpoints remain compatible. Export, PrivacyAction/RetentionHold, import recovery/status and privileged diagnostics may extend the catalogue.
Sprint 4 Error Catalogue	PASS WITH CONTROLLED ADDITIONS	Remains authoritative; Sprint 6 examples must be normalized/deduplicated before implementation.
Sprint 5 Security	PASS	No Sprint 6 decision weakens the accepted security baseline; implementation contracts realize its controls.
Authorization	PASS	PrincipalContext + current Event membership/global capability + lifecycle + resource policy + operation policy.
Transactions	PASS	Application-service ownership, no external side effects inside DB transactions, explicit retry eligibility.
Audit	PASS	Required audit atomically protects material mutations; typed payloads and tamper-evident versioned chains.
Jobs	PASS	Durable typed at-least-once jobs with leases, dedupe, bounded retry, dead-letter and reconciliation.
Privacy/Recovery	PASS	Archive/uninstall/user deletion are not purge; privacy tombstones and post-restore readiness reconciliation required.
Dependency graph	PASS	Potential cycles resolved through narrow ports/orchestrators; optimized projections remain read-only.

8. Controlled Extension Register

The following are forward implementation candidates. They must be introduced through reviewed versioned migrations or controlled catalogue/API updates; the frozen Sprint 3-5 baselines are not edited retroactively.

Surface	Extension family	Disposition
Database	Guest session/link, RSVP revision, durable Jobs/idempotency, recommendation/check-in metadata, Export/Privacy/Audit-integrity support	Versioned migrations after schema design review.
API	Export, PrivacyAction/RetentionHold, selected import recovery and sanitized diagnostics/status resources	Controlled additions to Sprint 4 endpoint catalogue; no breaking route rewrite.
Error/Audit/Capability catalogues	Normalize Sprint 6 codes/actions/capabilities into the existing authoritative Sprint 4/Sprint 5 catalogues	Deduplicate terminology before code implementation.

9. Implementation Gate

SVC-001 through SVC-016 are 16/16 design PASS; IV-001 through IV-020 are 20/20 design PASS; Sprint 3 database reconciliation is PASS with controlled migrations identified; Sprint 4 API reconciliation is PASS with controlled additions identified; Sprint 5 security reconciliation is PASS; blocking contradictions are NONE. EF-DOC-008 v1.0 is the approved Sprint 6 Application Services & Domain Implementation Design baseline. Production implementation may now proceed from this baseline, subject to controlled migrations/catalogue additions identified in this package. Implementation must execute the defined integration/security tests; a design PASS is an acceptance contract, not evidence that code has already passed runtime tests. Any later change to this baseline must be introduced through a controlled architecture decision and document revision.

Design validation defines implementation acceptance contracts; it does not claim that runtime code already exists or has passed implementation tests.